

Unit-1 Introduction to Disasters

Presented by
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Topics to be covered

- Definitions of Disaster, Hazard, Vulnerability, Risk, Exposure, Coping Capacity
- Risk Evaluation
- Disasters-Types, Causes, Consequences and Control
 - Geological Disasters,
 - Hydro-Meteorological Disasters,
 - Biological Disasters,
 - Technological Disasters,
 - Man-made Disasters
- Emerging Risks of Disasters – Climate Change and Urban Disasters

Objectives:

After studying this unit you should be able to

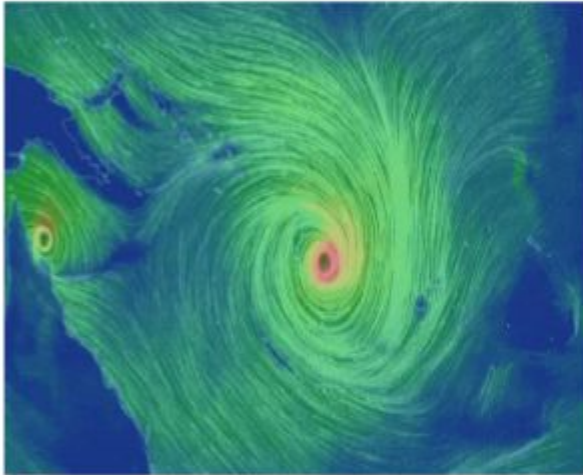
- Describe the various elements of disasters.
- Describe various types of disasters
- Explain the causes of disasters; and
- Discuss the effects of disasters

Worried about disaster



Disaster

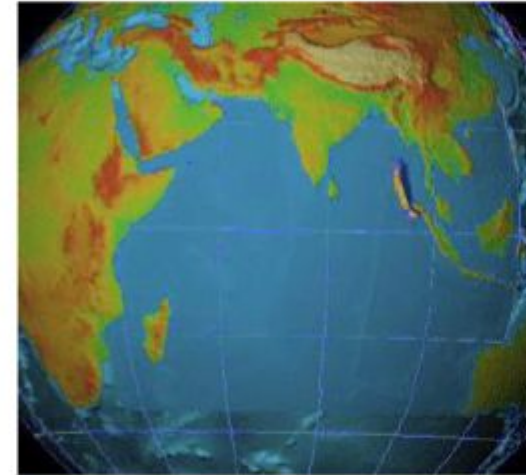
Disasters of 21st Century



1999 Super Cyclone
Orissa



2001 Bhuj Earthquake
Gujarat



2004 Tsunami
Southern India



2013 Flash Floods
Uttarakhand

What is Disaster?



Loss of Life (Social Loss)



**Damage to Property
(Economic Loss)**



What is Disaster?

- *Definition:* A disaster is a sudden, catastrophic event that causes significant disruption, destruction, and distress, often overwhelming the affected community's ability to cope.
- *Characteristics:*
 - **Sudden Impact:** Disasters occur rapidly, with little to no warning.
 - **Widespread Damage:** Affects a large area or population.
 - **Human and Economic Loss:** Results in the loss of life, injury, and economic damage.

To Understand the Disaster Risk some Terminologies need to be understood like :
Hazard, Vulnerability, Exposure and Coping Capacity.

Hazard

- *Definition:* A hazard is a source of potential harm or danger that may lead to a disaster. Hazards can be natural (e.g., earthquakes, floods) or human-made (e.g., industrial accidents, pollution).
- *Constituents:*
 - **Natural Hazards:** Result from natural processes.
 - **Human-made Hazards:** Arise from human activities.



Cyclone



Earthquake



Floods



Nuclear Accidents



Oil Spill



Explosions



Deforestation



Accidents



Pollution

<https://sciendtech.com/types-of-man-made-disasters/>

Vulnerability

- *Definition:* Vulnerability refers to the susceptibility of a community or system to the impacts of hazards. It is influenced by various factors, including social, economic, and environmental aspects.
- *Factors Influencing Vulnerability:*
 - **Poverty:** Lack of resources to cope with disasters.
 - **Population Density:** Higher density may increase vulnerability.
 - **Infrastructure Quality:** Well-built infrastructure can reduce vulnerability

Vulnerability is a characteristic of those 'assets' which are exposed!





Poorly planned urban development

Poverty as a Risk Driver

<https://www.youtube.com/watch?v=TH3LQW8B2Zw&t=54s>

Factors affecting vulnerability

$$\underline{\text{Vulnerability} = (\text{E} \times \text{S}) / \text{R}}$$

E=Exposure

S=Susceptibility (The state of being at risk, if exposed to a hazard”

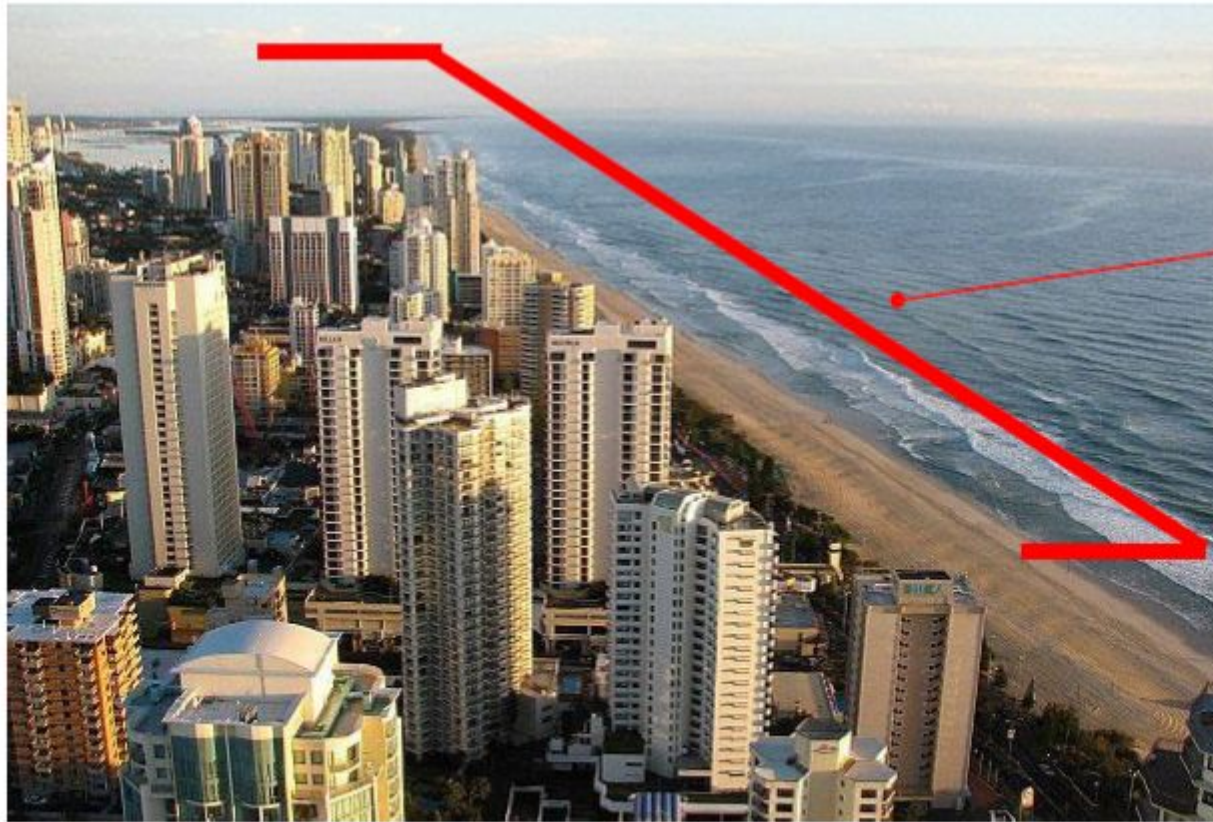
Example of susceptibility : Not being able to swim



R= Resilience (The ability of a system, community or society exposed to hazards to resist, absorb, accommodate to and recover from the effects of a hazard in a timely and efficient manner) EX: Living among people that can help you

What is Exposure?

- The situation of people, infrastructure, housing, production capacities and other tangible human assets located in hazard-prone areas.



Exposed!

https://www.youtube.com/watch?v=tkc3n8_wqps

Exposure

• *Just answer a simple question-*

When a hazard strikes, ‘what all**’ will it affect?**

This has two dimensions to it:



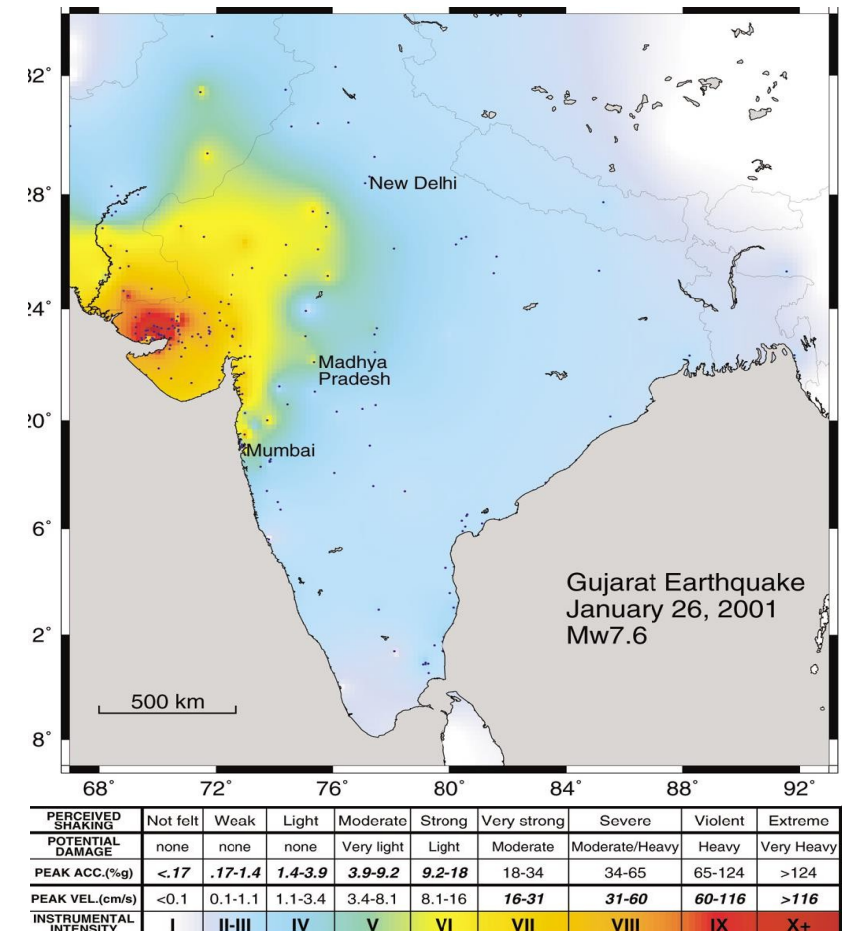
1. Location



2. Elements at the exposed location

EXPOSURE - Location

- **Location** is the area where the hazard will strike or strikes, and also the area where the effects of the hazard can be witnesses, experienced and felt.
 - Let's understand this with an earthquake example.
Earthquake is a hazard and we generally tend to **associate a location with an earthquake** like Kutch Bhuj in 2001 case.
 - We may feel that since the earthquake was caused due to a fault close to Bhuj (Bhachau), the exposure was primarily of Bhuj and adjoining areas, but interestingly the tremors were not restricted to Bhuj only.
-
- | PERCEIVED SHAKING | Not felt | Weak | Light | Moderate | Strong | Very strong | Severe | Violent | Extreme |
|-------------------|----------|---------|---------|------------|--------|-------------|----------------|---------|------------|
| POTENTIAL DAMAGE | none | none | none | Very light | Light | Moderate | Moderate/Heavy | Heavy | Very Heavy |
| PEAK ACC. (%g) | <.17 | .17-1.4 | 1.4-3.9 | 3.9-9.2 | 9.2-18 | 18-34 | 34-65 | 65-124 | >124 |



PERCEIVED SHAKING	Not felt	Weak	Light	Moderate	Strong	Very strong	Severe	Violent	Extreme
POTENTIAL DAMAGE	none	none	none	Very light	Light	Moderate	Moderate/Heavy	Heavy	Very Heavy
PEAK ACC.(%)	<1.7	1.7-1.4	1.4-3.9	3.9-9.2	9.2-18	18-34	34-65	65-124	>124
PEAK VEL.(cm/s)	<0.1	0.1-1.1	1.1-3.4	3.4-8.1	8.1-16	16-31	31-60	60-116	>116
INSTRUMENTAL INTENSITY	I	II-III	IV	V	VI	VII	VIII	IX	X+



Damage to Jaisalmer fort. Damage reported from Mt. Abu, Pokhran, Jodhpur, Jaipur & Udaipur

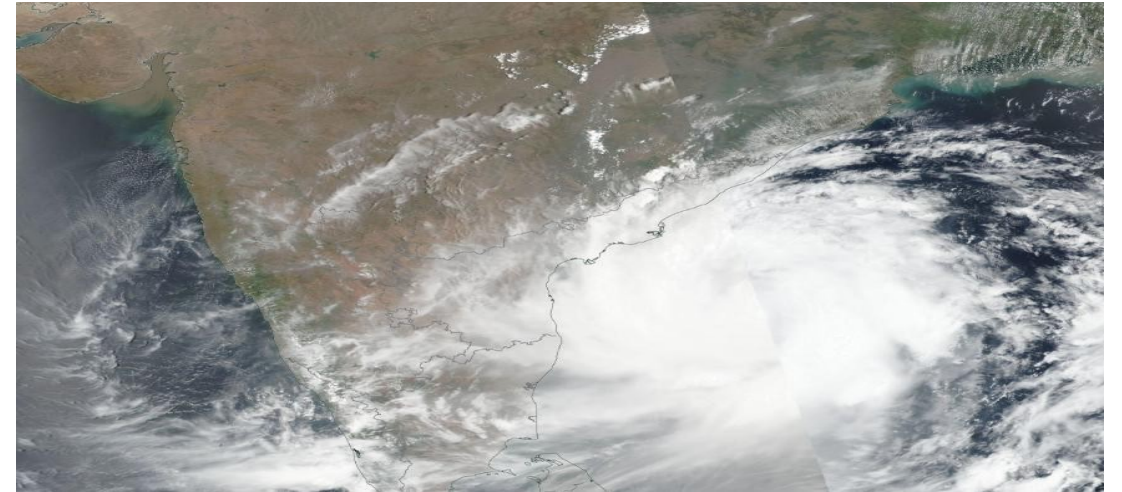
Tremors felt as far as Jabalpur

Building cracks in Mumbai & Vashi. Tremors felt as far as Kolhapur

Shock was also felt to a limited extent in Hyderabad & Bengaluru, and in high-rise buildings as far as Kolkata & New Delhi

Ex:2 Cyclone Fani of 2019.

- Coast of Odisha was the area primarily exposed to the cyclone, but, turbulent winds(280 km/h) was felt even in the interiors. If this cyclone would have been less intense, the coasts would have been affected and the interiors would have been affected much lesser.



<https://www.youtube.com/watch?v=exMSmXqEKKo>

- What is exposure in case of COVID 19 pandemic?



Search by Country, Territory, or Area

WHO Coronavirus Disease (COVID-19) Dashboard

Data last updated: 2020/5/14, 7:10pm CEST

Overview

Explorer



Globally, as of **7:10pm CEST, 14 May 2020**, there have been **4,258,666 confirmed cases** of COVID-19, including **294,190 deaths**, reported to WHO.



EXPOSURE - Elements at the exposed location



'Concern' in DRM is defined by damages, losses, etc.



No impact or effect
of Hazard

The concept of exposure is not only limited to the area, it also involves **elements like people, buildings/ housings, infrastructure (dams, canals, data-centres, distribution of optical fibre cables, communication towers, industries, refineries, etc.)**

Hazard impact on all such elements should be of Concern

Exposure is about People



Infrastructure



Buildings



- Measures of exposure can include the **number of people** or **types of assets** in an area.
- Let's say that there is an imminent danger of a forest fire, in that case the exposure would be of the forest area and the flora and fauna of the area!
- To put it simply, if you feel that you can be affected by a hazard, you are exposed!



Think



Should an earthquake here be a concern to us?



Risk

- *Definition:* Risk is the likelihood of a hazard causing harm and the magnitude of that harm. It involves the probability of an event occurring and its consequences.
- *Components:*
 - **Hazard Identification:** Recognizing potential threats.
 - **Vulnerability Assessment:** Evaluating the susceptibility to harm.
 - **Risk Assessment:** Determining the likelihood and impact of an event.

Risk Assessment Matrix

It's a tool that helps you visualize the probability versus the severity of a potential risk.

		Consequences				
		Insignificant (1) No injuries / minimal financial loss	Minor (2) First aid treatment / medium financial loss	Moderate (3) Medical treatment / high financial loss	Major (4) Hospitalable / large financial loss	Catastrophic (5) Death / massive financial loss
Likelihood	Almost Certain (5) Often occurs / once a week	Moderate (5)	High (10)	High (15)	Catastrophic (20)	Catastrophic (25)
	Likely (4) Could easily happen / once a month	Moderate (4)	Moderate (8)	High (12)	Catastrophic (16)	Catastrophic (20)
	Possible (3) Could happen or known it to happen / once a year	Low (3)	Moderate (6)	Moderate (9)	High (12)	High (15)
	Unlikely (2) Hasn't happened yet but could / once every 10 years	Low (2)	Moderate (4)	Moderate (6)	Moderate (8)	High (10)
	Rare (1) Conceivable but only on extreme circumstances / once in 100 years	Low (1)	Low (2)	Low (3)	Moderate (4)	Moderate (5)

Depending on likelihood and severity, risks can be categorized as high, moderate, or low. As part of the [risk management process](#), companies use risk matrices to help them prioritize different risks and develop an appropriate mitigation strategy.

<https://www.safeworkpro.com.au/an-example-of-risk-assessment-matrix>

How to Determine the Likelihood of a Risk Occurring

- **5: Highly Likely.** Risks in the highly likely category are almost certain to occur. Typically, risks with **91 percent** or more likelihood fall into this category.
- **4: Likely.** A likely risk has a **61-90** percent chance of occurring. These risks need regular attention, as they are bound to reoccur and therefore require a consistent mitigation strategy.
- **3: Possible.** Possible risks may happen about half the time — they have a **41-60 percent** chance of occurring and need attention.
- **2: Unlikely.** Risks in the unlikely category have a relatively low chance of occurring — **11 to 40 percent**. But they may still affect your business, so it's a good idea to keep an eye on them.
- **1: Highly Unlikely.** Highly unlikely risks are exactly as they sound, with a low probability of occurring.

5x5 Risk Matrix

Severity —

		1 Insignificant	2 Minor	3 Moderate	4 Major	5 Death
Likelihood ↑	1 Rare	1	2	3	4	5
	2 Unlikely	2	4	6	8	10
	3 Possible	3	6	9	12	15
	4 Likely	4	8	12	16	20
	5 Certain	5	10	15	20	25

How Does a Risk Matrix Work?

- Risks come in many forms: strategic, operational, financial, and external. The risk assessment matrix works by presenting various risks as a chart, color-coded by severity: high risks in red, moderate risks in yellow, and low risks in green. Every risk matrix also has two axes: one measuring likelihood and one measuring impact.
- Likely risk events may have a 61 to 90 percent chance of occurring, while highly unlikely events are extremely rare, with a less than 10 percent chance of occurring. Depending on the business and its risk appetite, an insignificant impact may cause a negligible amount of damage — such as a loss of less than \$1K — while a catastrophic impact might create losses of \$1M or more.
- By grading the risk event's likelihood and impact, the risk matrix provides a quick snapshot of the threat landscape. Visualizing the threat landscape in this way, audit, risk, and compliance professionals can more easily foresee and determine how to minimize events that can have a substantial impact on the company.

Why Is a Risk Matrix Important?

- A risk matrix can help businesses cultivate a solid understanding of the risk environment, helping them manage and mitigate risks before they occur.

HOW CAN WE REDUCE RISK?

$$\text{RISK} = \text{HAZARD} \times \text{EXPOSURE} \times \text{VULNERABILITY}$$



We can improve our abilities to monitor and forecast hazards



Increased awareness of the hazards faced by communities and their exposure to them



The greatest benefits can be achieved by reducing the vulnerability to natural hazards

Coping Capacity

- **Definition:** Coping Capacity in disaster management refers to the ability of individuals, communities, and systems to effectively prepare for, respond to, and recover from disasters.
- **Components:**
 - **Individual Capacity:** Knowledge and skills of individuals.
 - **Community Capacity:** Community organization and resources.
 - **Institutional Capacity:** Government and organizational capabilities.

What is Disaster Risk?

- Now that we know few of the important terms, let us try to understand what a disaster risk is.
- **Hazard**, which can be natural or man-made, will occur at some point of time, whether we want it or not. So basically, it is a probability of occurrence.
- Given this probability of occurrence of a hazard, there will be ‘something’ which will be **exposed** to the impending effects.
- Now this exposed assets will have **vulnerabilities** to different extents. Some may be highly vulnerable, while, some will be very less vulnerable.

$$\text{Disaster Risk} \propto \frac{\text{Hazard} \times \text{Exposure} \times \text{Vulnerability}}{\text{Coping Capacity}}$$

Understanding of Disaster Risk with different cases

- Case 1: We often hear about avalanches in the upper reaches of Himalayas, but we never call it a disaster. Why? Because, there is an imminent hazard but nothing is exposed and if nothing is exposed, there will be no vulnerabilities.



- Case 2: Let's assume another scenario. We build a strong house in an earthquake prone area and there is an earthquake. Nothing happens to the building or the people living inside it. In this case, there is a hazard and the building is exposed, but, the building has very less or no vulnerability at all.

PROTECT YOUR HOME FROM AN EARTHQUAKE



- Case 3: In the same scenario, let's say that although the building is strong but the **people inside are not aware of what to do in an earthquake**. So, all 200 people inside the building start running out.
- This creates a stampede and when they come out, other building outside, which are not so strong, collapses causing injury and death. In this case, the vulnerability of the people due to the lack of knowledge raises the disaster risk.



Learning Line

If you have the capacity to deal with the disaster risk, you are good to go, else it is a disaster



Think of it this way.

- In case of hazards like earthquake or cyclone or tsunami, they are all natural phenomenon which can not be stopped. However, knowingly we build our houses in hazard prone zones and above that, we do not even take care of our vulnerabilities to increase our coping capacities. So, we, humans, are actually increasing the disaster risks to an extent where it is becoming a disaster. **So, are these disasters really are natural or are we responsible for it in some ways?**
- In case of hazards like oil-spills and refinery explosions, it is off course, the humans who are responsible. **To sum it up, are there any disasters which are natural?**

- Bhopal Gas Tragedy video to relate factors of Disaster Risk:
<https://youtu.be/YPvktXuc55A?si=wn8HEQzYM1m0aB2K>

Types of Hazards/Disaster

- Natural and manmade Hazards have been classified into different ‘scientific categories’ depending on how they occur.



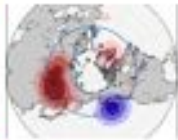
Geophysical

Hydrological



Meteorological

Climatological



Biological

Technological



Geophysical: Hazards originating from internal earth processes. Eg. earthquakes, tsunami, volcanic activity

Hydrological: Events caused by deviations in the normal water cycle and/or overflow of bodies of water. Eg. Floods, coastal erosion, landslides

Meteorological: Events caused by micro to meso-scale atmospheric processes and extreme weather conditions (in the spectrum from minutes to days).
Eg. Cyclone, heatwaves, lightning

Climatological: Unusual, extreme weather conditions caused by long-lived meso to macro- scale processes (in the spectrum from intra-seasonal to multi-decadal climate variability). Eg. Drought, wildfires, GLOF(Glacial Lake Outburst Flood)

Biological: Process or phenomenon of organic origin or conveyed by biological vectors, including exposure to pathogenic micro-organisms, toxins and bioactive substances. Eg. Epidemics, insect infestation

Technological: Originate from technological or industrial conditions, dangerous procedures, infrastructure failures or specific human activities. E.g. nuclear radiation, dam failures, factory explosions, fires and chemical spills.

Identify The Type of Hazard/ Disaster



Geological



Geological



Biological



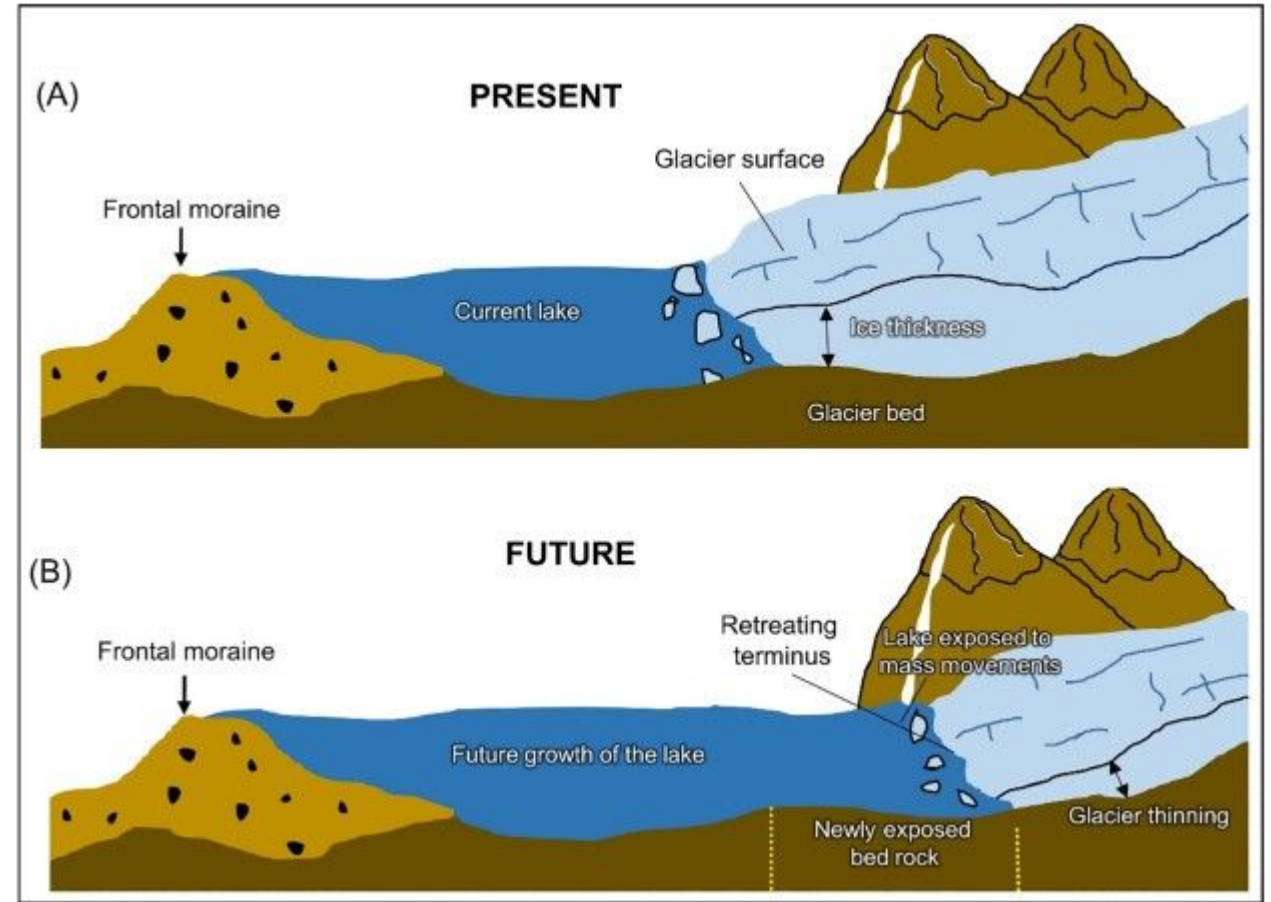
Hydrological



Coastal Erosion- Hydrological



Wild Fire- climatological



GLOF-climatological

<https://www.drishtiias.com/daily-updates/daily-news-analysis/glacial-lake-outburst-flood-in-sikkim#:~:text=Sikkim%20recently%20experienced%20a%20Glacial,a%20result%20of%20continuous%20rainfall>



Technological



Disaster Trend

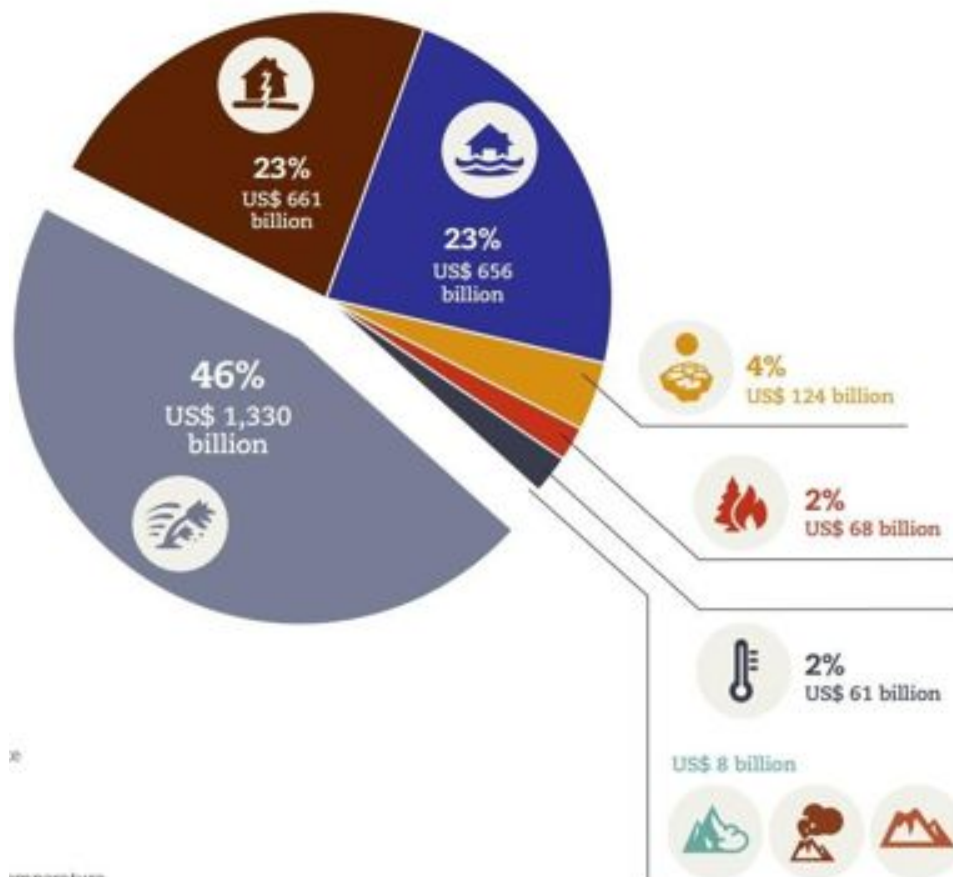
Why do we need to worry? Occurrences (1998-2017)



Effect of Disaster

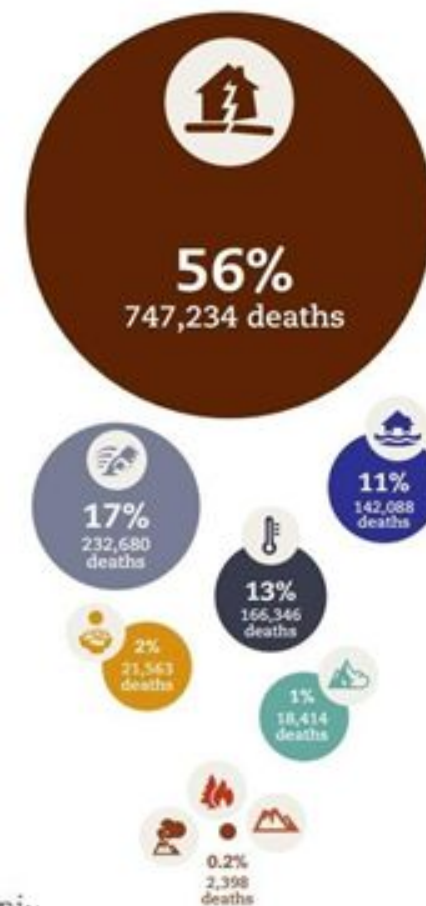
सत्यमेव जयते

Economic Losses



Temperature
Landslide, Volcanic activity, Mass movement

Deaths



- Storm
- Earthquake
- Flood
- Drought
- Wildfire
- Extreme temperature
- Others : Landslide, Volcanic activity, Mass movement

Effect of disaster in India (1998 – 2017)



13 lakh Killed



440 crore Injured, Homeless, Displaced



US\$ 2908 Bn. Direct Economic Loss

Report to study on trends in disaster :

https://www.ifrc.org/sites/default/files/2023-01/20230130_2022_WDR_DataAnnex.pdf

References :

https://nidm.gov.in/pdf/disaster_trends.pdf

<https://egyankosh.ac.in/bitstream/123456789/26452/4/Unit-2.pdf>

<https://www.youtube.com/watch?v=NJixgVxTB7I>

Kutch Earthquake Video

https://www.youtube.com/watch?v=xBV_Z1meGNA